Mini Project Report

A report on

AIRLINE FLIGHT AND PRICE ANALYTICS DASHBOARD

Submitted to

Data Visualization
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In partial fulfillment of the requirements for the Mini Project

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1. INTRODUCTION

Air travel is an important part of modern transportation. Flight ticket prices can vary depending on the airline, travel class, source city, destination city, number of stops, flight duration and the number of days left before travel. Looking at these details in a raw spreadsheet can make comparisons difficult.

The Airline Flight and Price Analytics Dashboard is a mini project developed using Microsoft Power BI Desktop. It converts flight-ticket records into visual summaries such as KPI cards, charts and filters. The dashboard helps users compare ticket prices and explore flight information by airline, class, city and other categories.

2. PROJECT OBJECTIVE

- To analyze airline flight and ticket-price data using Microsoft Power BI.
- To summarize total flight records, number of airlines, average ticket price and average flight duration.
- To compare ticket prices across airlines, travel classes and flight categories.
- To use charts and slicers to make the dashboard interactive and easy to understand.
- To present findings clearly and consider the limitations of the dataset.

3. TOOLS AND TECHNOLOGIES

- Microsoft Power BI Desktop – dashboard creation and visualization.
- Power Query – data cleaning and transformation.
- CSV flight dataset – source of airline and ticket-price information.
- Power BI visuals – KPI cards, donut chart, treemap, bar/column chart, line chart and slicers.

4. PROJECT WORKFLOW

The project follows a data-visualization workflow. Each step helps prepare the data and select visuals that answer the main questions.

1	Select Real Dataset
2	Understand the Data
3	Clean the Data
4	Define the Question
5	Explore the Data
6	Select Important KPIs
7	Choose Appropriate Charts
8	Build Dashboard
9	Create Narrative
10	Evaluate for Bias
11	Present Findings

5. SELECT REAL DATASET

An Indian flight-ticket dataset was selected for this project. It contains flight-related details such as airline, flight identifier, source city, departure time, number of stops, arrival time, destination city, travel class, duration, days left and price. The dataset is suitable for comparing ticket prices and examining differences between flight categories.

6. UNDERSTAND THE DATA

Field	Description	Type
Airline	Name of the airline	Category
Flight	Flight identifier or number	Text
Source City	City from which the flight departs	Category
Departure Time	Departure-time category	Category
Stops	Number/category of stops	Category
Arrival Time	Arrival-time category	Category
Destination City	Flight destination city	Category
Class	Travel class, e.g. Business or Economy	Category
Duration	Flight duration	Numeric/time
Days Left	Days remaining before the flight date	Number
Price	Ticket price	Number/currency

7. CLEAN THE DATA

The dataset was checked in Power Query before building visuals. The following checks should be completed and retained as part of the project workflow:

• Check for blank or missing values in important columns.

• Check and remove duplicate records only when they are true duplicates.

• Confirm that Price, Days Left and other numeric fields use suitable data types.

• Confirm that Duration is consistently represented and can be used correctly in visuals.

• Check text fields for inconsistent spelling or extra spaces.

• Format ticket price as currency for display, while keeping the numeric data type.

8. DEFINE THE QUESTION

The main question for this project is: How do ticket prices vary across airlines, travel classes and flight details?

The dashboard is designed to explore the following questions:

• How many flight records are available in the dataset?

• How many different airlines are represented?

• What is the average ticket price?

• What is the average flight duration?

• How do ticket prices differ across airline categories?

• How do Business and Economy class prices compare?

• How are flight records distributed across the categories shown in the dashboard?

• How can users filter the dashboard by airline, city, class or stops?

9. EXPLORE THE DATA

The data was explored using simple charts and summaries before finalizing the dashboard. Average ticket price was compared across categories, and flight records were examined by airline and travel class. The visuals were selected to make comparisons easy to understand without reading every row of the dataset.

The exploration stage helps identify which fields are useful for grouping, which fields can be measured numerically, and which filters will make the dashboard more useful.

10. SELECT IMPORTANT KPIs

Key Performance Indicators (KPIs) provide a quick summary at the top of the dashboard. The selected KPI cards are:

KPI	Meaning	Calculation / Field
Total Flights	Number of flight records in the loaded dataset	Count of flight records
Number of Airlines	Number of distinct airline names	Distinct count of Airline
Average Ticket Price	Typical ticket price across records	Average of Price
Average Flight Time	Typical flight duration	Average of Duration, if stored numerically

KPI values should be read directly from the final Power BI report after all filters and calculations are confirmed. The values may change when a slicer is selected.

Note: If Duration is stored as text, it must be transformed into a suitable numeric duration measure before calculating an average. Do not calculate an average directly from text values.

11. CHOOSE APPROPRIATE CHARTS

Visual	Used For	Question Answered
KPI Cards	Show key summary values	What is the overall flight-data summary?
Donut Chart	Show category distribution	How are records distributed across selected categories?
Treemap	Compare average price across categories	Which categories show different average prices?
Bar/Column Chart	Compare values between categories	How do selected flight categories compare?
Line Chart	Show change across an ordered field	How does average price vary across the selected sequence?
Slicers	Filter by airline, city, class or stops	How does the result change for a selected group?

12. BUILD DASHBOARD

The dashboard was created in Power BI Desktop. The title, KPI cards and charts were arranged to make the main information visible first. Different colors were used to separate the KPI cards and categories. Filters can be added for Airline, Source City, Destination City, Class and Stops, depending on the fields available in the loaded dataset.

The report is interactive: selecting a slicer value changes the displayed results in the connected visuals. The design should remain uncluttered, with readable titles, consistent colors and enough space between charts.

13. CREATE NARRATIVE

The dashboard tells a simple story. First, the KPI cards provide a quick overview of flight records, airlines, average ticket price and average flight duration. Next, the charts allow users to compare categories and inspect differences in ticket prices. Finally, slicers let users focus on a particular airline, city, travel class or stop category.

The purpose is not only to display numbers, but also to help users explore the data and understand how ticket prices vary across the available categories.

14. EVALUATE FOR BIAS

• Use the available dataset consistently and do not invent missing values or findings.

• Remember that the dataset represents a particular collection of flight records and may not represent every airline, route or date.

• Ticket prices can depend on travel date, booking time, route, class and other factors that may not all be present in the dataset.

• A difference between two categories does not by itself prove that one factor caused the price difference.

• Use clear chart titles and readable scales so that comparisons are not misleading.

• Describe this as analysis of a real-world dataset; do not call it live real-time data unless the data is connected to a live source.

15. OUTPUT

The following pages are reserved for screenshots of the actual Airline Flight and Price Analytics Dashboard created in Power BI. Replace each placeholder with a screenshot from your own final dashboard before submitting the report.

Figure 1: Airline Flight and Price Analytics Dashboard – Complete Overview

Paste your actual Power BI screenshot here.

Figure 1: Complete dashboard overview

15. OUTPUT (CONTINUED)

Figure 2: KPI Cards – Total Flights, Number of Airlines, Average Ticket Price and Average Flight Time

Paste your actual Power BI screenshot here.

Figure 2: KPI cards used for the summary

Figure 3: Price Comparison Chart / Treemap

Paste your actual Power BI screenshot here.

Figure 3: Price comparison across categories

15. OUTPUT (CONTINUED)

Figure 4: Donut Chart and Category Distribution

Paste your actual Power BI screenshot here.

Figure 4: Distribution of flight records across categories

Figure 5: Bar Chart and Line Chart

Paste your actual Power BI screenshot here.

Figure 5: Comparison and trend visuals

15. OUTPUT (CONTINUED)

Figure 6: Slicers / Filters – Airline, Source City, Destination City, Class and Stops

Paste your actual Power BI screenshot here.

Figure 6: Dashboard filters

Before submission, ensure the screenshots show the actual Power BI visuals, the correct dashboard title and the final filter layout. Do not leave these placeholder boxes in the final printed copy.

16. PRESENT FINDINGS

The completed dashboard is intended to summarize flight-ticket information and help users compare airline-related categories. The KPI cards give a quick overview of the loaded data, while the charts show differences across the selected fields. Slicers allow users to investigate a particular airline, city, class or stop category without creating a separate report.

The final numerical findings should be recorded from the actual Power BI dashboard after checking the dataset and confirming the measures. Examples include the total number of loaded flight records, the number of airlines, the average ticket price and the average duration. These values may change when filters are applied.

CONCLUSION

The Airline Flight and Price Analytics Dashboard demonstrates how a flight-ticket dataset can be converted into an interactive report using Microsoft Power BI. By selecting and understanding the dataset, cleaning the data, defining questions, choosing KPIs and charts, building the dashboard and reviewing possible bias, the project presents information in a clear visual format. It helps users explore airline data and compare ticket prices across available categories.

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